Guarded summation-polynomial fall degrees

Study report, 28 October 2026. Derived bounds and development checks; no performance claim.

Let S be B distinct rational affine-point abscissas, $g(T) = \prod_{s\in S}(T - s)$, and $r = x(R)$ for an affine rational target. The ordinary polynomial-ring inputs are

$$g(x), \quad g(y), \quad g(z), \quad S_3(x, y, u), \quad S_3(u, z, r), \quad v(x - y) - 1.$$

Both point signs are allowed. The guard requires distinct first abscissas. Every degree below is ordinary total degree with explicit membership equations. The first-fall convention is homogeneous-top syzygies modulo Koszul syzygies.

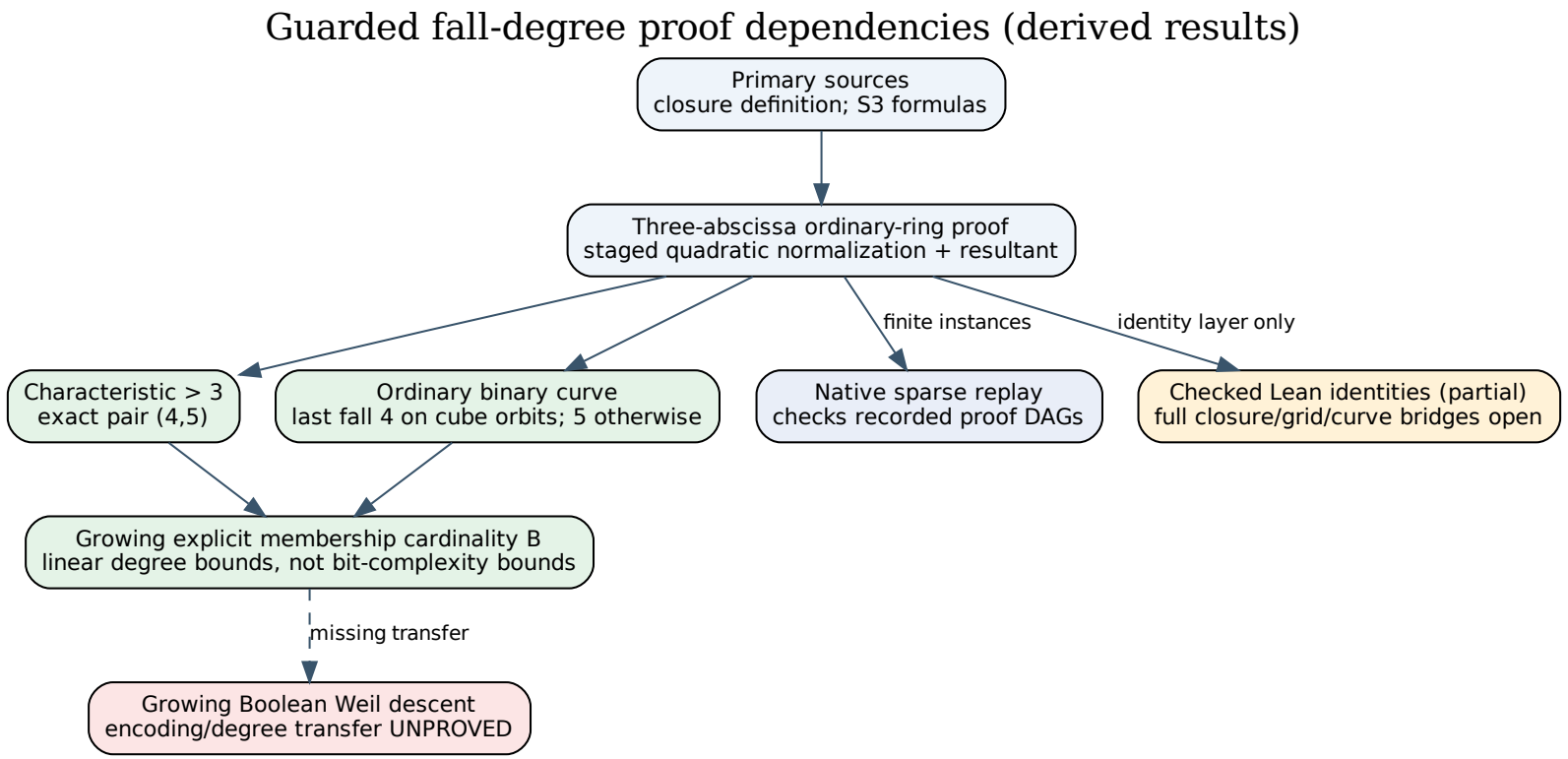
Setting, unsatisfiable instances	First fall	Last fall
Short Weierstrass, characteristic > 3, $B = 3$	4	5
Same model, $B \geq 3$	$B + 1$	$\max(5, B + 1) \leq d_F \leq 3B - 4$
Ordinary binary, $B = 3$, $g(T) = T^3 + c$	4	4
Ordinary binary, $B = 3$, other split cubics	4	5
Ordinary binary, $B \geq 4$	$B + 1$	$B + 1 \leq d_F \leq 3B - 4$

The binary curve is $Y^2 + XY = X^3 + a_2X^2 + a_6$, $a_6 \neq 0$. Its cube-orbit exception requires even extension degree, so the pair is $(4, 5)$ for all three-abscissa unsatisfiable instances over $\mathbb{F}_{2^n}$ with odd n. The bounds are mathematical derivations, not empirical fits. The growing lower bound depends on the degree-B membership input and gives no corresponding lower bound in the variable count of a fixed-degree Boolean encoding.

Method. Staged multiplication and cubic reduction normalize two quadratic fibres, then derive their scalar resultant within degree five. Finite-grid inversion yields 1. Characteristic two uses a Frobenius identity instead of division by two. Closed separating superspaces give the lower bounds. An exact triple-selector partition lifts local proofs to growing cardinality.

Verification boundary. Native sparse replay checks the compact historical proof DAGs against separately supplied inputs. Native binary-field arithmetic and the pinned cube-orbit fixture have development unit checks; no new field/base/target sweep or timing measurement was run. Lean checks only the declarations and hypotheses listed in `formal/PROOF_MAP.json`. The complete filtered-closure, grid-inversion, and elliptic-curve bridges are not formalized in this packet.

Prior work. The checked sources are Huang–Kosters–Yang–Yeo, 1505.02532; Caminata–Gorla, 2112.05579; Semaev, 1504.01175; and Kosters–Yeo, 1503.08001. Assumptions and repository links are in `SOURCES.md`. No literature-priority claim is made.



Solid arrows denote dependencies in the written arguments. The dashed arrow is an unresolved encoding/degree transfer, not a proved implication. The Lean box certifies algebraic identity lemmas only; native replay certifies finite proof operations only. Neither substitutes for the complete mathematical derivation.

The existing E12 Boolean Weil-descent records are unchanged. Its completed cell retains labels $\geq 5, 4, \geq 5, 4$, a frozen lower median of 4, and a separate independent closure-replay gap. The present coefficient-field theorems do not close that gap. No cost ratio, benchmark, leaderboard or security claim follows.

Reproducible records. See `README.md`, the proof files, `native/validation/`, `identities/`, and `formal/`. The editable diagram source is `reports/proof-dependencies.dot`. Historical Python sources are preserved as provenance, not as an execution path.